Synoptic Question 18: The Physics of a Wave Energy Converter

Reading Passage

Read the following article and then answer the questions that follow.

Ocean wave energy is a vast, untapped renewable resource. One common method for harnessing this energy is using a "point absorber" Wave Energy Converter (WEC). This device consists of a floating buoy on the surface of the ocean, tethered to a fixed reference point, such as a large spar or an anchor on the seabed.

As waves pass, they cause the buoy to move up and down (heave). This relative motion between the floating buoy and the fixed reference is used to drive a generator. In many designs, this motion compresses a fluid or drives a linear electrical generator directly, converting the wave's kinetic and potential energy into electrical energy.

The power captured by the device is highly dependent on the wave height and the wave period. The buoy is designed to oscillate in resonance with the incoming waves. Resonance occurs when the period of the waves is close to the natural period of oscillation of the buoy. This maximises the buoy's amplitude of motion and, therefore, its energy capture.

The energy in an ocean wave is a combination of the kinetic energy of the moving water and the gravitational potential energy of the raised water. The generator, by extracting this energy, creates a "damping" force on the buoy, which prevents its oscillations from becoming infinitely large.

Data:

Data for a specific location and a prototype WEC (see Fig. 18.1):

Table 1: Data for Wave Energy Converter

Parameter	**Value**
Average wave height (amplitude A), H	3.0 m
Average wave period, T	8.0 s
Speed of the waves, v	12.0 m s^{-1}
Density of seawater, ρ	1025 kg m^{-3}
Mass of the floating buoy, m	25,000 kg
Diameter of the buoy, d	6.0 m
Natural period of buoy's oscillation, T_{nat}	7.5 s
Average electrical power output, P_{elec}	45.0 kW
Resistance of undersea cable to shore, R_{cable}	2.2 Ω
Output voltage of WEC, V_{out}	4100 V

[Simple Diagram of a Point Absorber WEC]

Wave Direction →
Buoy (Oscillates Up and Down)
↑ *Motion* ↓
Spar/Column (Contains Generator)
Seabed

Figure 1: Schematic of a Point Absorber WEC

Questions

(a) First, consider the properties of the ocean waves.

(i) Calculate the frequency, f, of the waves.

(ii) Calculate the wavelength, λ, of the waves.

(b) The power available in a wave, P_{wave}, per metre of wavefront width is given by:

$$P_{wave} \approx \frac{1}{8}\rho g A^2 v$$

Note: The wave height $H = 3.0$ m given in the table is the amplitude A.

(i) Calculate the power available per metre of the wave's width.

(ii) The buoy has a diameter of 6.0 m. Estimate the total wave power $P_{incident}$ that is incident on the buoy.

(c) Using the $P_{incident}$ from (b)(ii) and the P_{elec} from the data, calculate the overall efficiency of the WEC.

(d) The buoy oscillates with the waves. Assume its motion is Simple Harmonic Motion (SHM) with an amplitude equal to the wave amplitude ($A = 3.0$ m) and a period equal to the wave period ($T = 8.0$ s).

(i) Calculate the maximum speed, v_{max}, of the buoy as it oscillates.

(ii) Calculate the maximum kinetic energy, KE_{max}, of the buoy.

(e) The buoy's design is based on resonance.

(i) Explain what is meant by resonance.

(ii) Compare the wave period to the buoy's natural period and comment on the effectiveness of the design.

(f) The generator that produces the electrical power also acts to slow the buoy down.

(i) What is the name given to the force that the generator exerts on the oscillating system?

(ii) Briefly state the main energy conversion taking place in the generator.

(g) The electrical power is transmitted to shore.

(i) Calculate the current I in the undersea cable.

(ii) Calculate the power lost as heat in the cable.

(iii) Calculate the total energy, in joules, produced by the WEC in one day.

Solutions

(a) (i) $f = \frac{1}{T} = \frac{1}{8.0} = 0.125$ Hz.

(ii) $v = f\lambda \implies \lambda = \frac{v}{f} = \frac{12.0}{0.125} = 96.0$ m.

(b) (i) $P_{wave} = \frac{1}{8}(1025)(9.81)(3.0)^2(12.0) = 1.36 \times 10^5$ W m^{-1}.

(ii) The buoy interacts with a width of wave equal to its diameter.
$P_{incident} = (1.36 \times 10^5 \text{ W m}^{-1}) \times (6.0 \text{ m}) = 8.16 \times 10^5$ W (or 816 kW).

(c) $\eta = \frac{P_{out}}{P_{in}} = \frac{P_{elec}}{P_{incident}} = \frac{45.0\times10^3}{8.16\times10^5} = 0.0551 = 5.5\%$.

(d) (i) $\omega = \frac{2\pi}{T} = \frac{2\pi}{8.0} = 0.785$ rad s^{-1}.
$v_{max} = \omega A = (0.785)(3.0) = 2.36$ m s^{-1}.

(ii) $KE_{max} = \frac{1}{2}mv_{max}^2 = \frac{1}{2}(25000)(2.36)^2 = 69620 \text{ J} \approx 6.96 \times 10^4$ J.

(e) (i) Resonance is the phenomenon where a system is driven by an external periodic force at a frequency (the driving frequency) that is close to the system's own natural frequency of oscillation.

(ii) The wave period ($T = 8.0$ s) is the period of the driving force. The buoy's natural period is $T_{nat} = 7.5$ s. Since these two periods are very close, the system is operating near resonance, which will cause a large amplitude of oscillation and maximise energy capture.

(f) (i) This is a **damping** force (specifically, the "power take-off" damping).

(ii) The generator converts the **kinetic energy** (from the buoy's motion) into **electrical energy** (via electromagnetic induction).

(g) (i) $P = VI \implies I = \frac{P_{elec}}{V_{out}} = \frac{45.0\times10^3}{4100} = 10.98 \text{ A} \approx 11.0$ A.

(ii) $P_{loss} = I^2R_{cable} = (10.98)^2(2.2) = 265$ W.

(iii) $t = 1$ day $= 24$ h $\times$ 3600 s/h $= 86400$ s.
$E = P_{elec} \times t = (45.0 \times 10^3) \times (86400) = 3.89 \times 10^9$ J.

CIE 9702 Physics Paper 4 (A Level)
Appended Material: Variant 2 (Higher Difficulty Only)

This section is appended after the original uploaded paper. It is written in the style of CIE 9702 Paper 4 with structured parts, open-ended conceptual prompts, mark allocations, sample solutions, and a detailed mark scheme.

Question Paper (Variant 2)

Question 18: Advanced analysis of a wave-energy converter (WEC)

Ocean wave energy is harvested at a test site using a point-absorber WEC. A floating buoy oscillates vertically and drives a generator through a power take-off (PTO) system.

For this variant, use the following data:

Parameter	Value
Wave amplitude, A	$3.4\,\mathrm{m}$
Wave period, T	$7.2\,\mathrm{s}$
Wave speed, v	$11.5\,\mathrm{ms^{-1}}$
Density of seawater, ρ	$1025\,\mathrm{kgm^{-3}}$
Acceleration due to gravity, g	$9.81\,\mathrm{ms^{-2}}$
Mass of buoy, m	$2.80 \times 10^4\,\mathrm{kg}$
Buoy diameter, d	$7.0\,\mathrm{m}$
Natural period of buoy, T_0	$7.0\,\mathrm{s}$
Measured electrical output power, P_{elec}	$62.0\,\mathrm{kW}$
Generator output voltage, V	$3.60\,\mathrm{kV}$
Cable resistance to shore, R	$1.8\,\Omega$

Assume, where needed, that average wave power per metre of crest is

$$P_{\mathrm{wave}} = \frac{1}{8}\rho g A^2 v.$$

(a) Calculate:

(i) the wave frequency,

(ii) the wavelength.

[4]

(b) Determine:

(i) the wave power per metre of crest,

(ii) the total incident power intercepted by the buoy, assuming interception width $\approx d$,

(iii) the overall conversion efficiency from incident wave power to electrical output.

[6]

(c) Model the buoy as SHM with amplitude A and period T.

(i) Calculate angular frequency ω and maximum speed $v_{\max}$.

(ii) Calculate maximum acceleration $a_{\max}$.

(iii) Show that the maximum kinetic energy of the buoy is about 1.2×10^5 J.

[7]

(d) The buoy and PTO can be approximated as a forced oscillator.

(i) Calculate the driving frequency and the natural frequency.

(ii) Comment quantitatively on how close the system is to resonance.

(iii) Explain, with physics reasoning, why operating too close to exact resonance may be undesirable in real sea states.

[7]

(e) Electrical transmission to shore:

(i) Calculate transmission current.

(ii) Calculate cable power loss.

(iii) Determine the percentage of generated electrical power lost in the cable.

(iv) State one engineering method to reduce transmission losses and explain why it works.

[7]

(f) Open-ended concept question (energy flow and limits):

Using principles of mechanics and electricity, discuss why the measured efficiency in this system is far below 100%. Your answer should include:

- at least three distinct physical loss mechanisms,
- a clear energy-flow pathway from wave motion to delivered shore power,
- one justified modification that could improve average output in irregular seas.

[9]

(g) Open-ended design question (evaluation):

At this location, wave period varies between 5.5 s and 9.0 s over a day. Evaluate whether the chosen natural period $T_0 = 7.0$ s is a good design compromise. Support your argument with relevant resonance ideas and at least one quantitative comparison. [10]

[Total: 50]

Sample Solutions (Variant 2)

(a)

$$f = \frac{1}{T} = \frac{1}{7.2} = 0.1389\,\text{Hz} \approx 0.139\,\text{Hz}$$
$$\lambda = vT = (11.5)(7.2) = 82.8\,\text{m}$$

(b)

$$P_{\text{wave}} = \frac{1}{8}(1025)(9.81)(3.4)^2(11.5) = 1.67 \times 10^5\,\text{Wm}^{-1}$$
$$P_{\text{inc}} = P_{\text{wave}}d = (1.67 \times 10^5)(7.0) = 1.17 \times 10^6\,\text{W}$$
$$\eta = \frac{P_{\text{elec}}}{P_{\text{inc}}} = \frac{62.0 \times 10^3}{1.17 \times 10^6} = 0.053 \approx 5.3\%$$

(c)

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{7.2} = 0.873\,\text{rads}^{-1}$$
$$v_{\max} = \omega A = (0.873)(3.4) = 2.97\,\text{ms}^{-1}$$
$$a_{\max} = \omega^2 A = (0.873)^2(3.4) = 2.59\,\text{ms}^{-2}$$
$$KE_{\max} = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}(2.80 \times 10^4)(2.97)^2 = 1.23 \times 10^5\,\text{J}$$

(d) Driving frequency:

$$f_d = \frac{1}{7.2} = 0.1389\,\text{Hz}$$

Natural frequency:

$$f_0 = \frac{1}{7.0} = 0.1429\,\text{Hz}$$

Difference:

$$\Delta f = |f_0 - f_d| = 0.0040\,\text{Hz}, \quad \frac{\Delta f}{f_0} \approx 2.8\%$$

So the system is close to resonance. Exact/near-exact resonance can produce very large motion amplitudes, causing structural fatigue, end-stop impacts, and unstable PTO loading in irregular wave trains.

(e)

$$I = \frac{P}{V} = \frac{62.0 \times 10^3}{3.60 \times 10^3} = 17.2\,\text{A}$$
$$P_{\text{loss}} = I^2R = (17.2)^2(1.8) = 5.34 \times 10^2\,\text{W} = 0.534\,\text{kW}$$
$$\%\ \text{loss} = \frac{0.534}{62.0} \times 100 = 0.86\%$$

One method: increase transmission voltage (for same power this lowers current), reducing I^2R heating losses.

(f) Suggested high-quality response points

Energy flow: wave mechanical energy → buoy oscillation → PTO mechanical transfer → generator electrical output → cable delivery to shore.

Reasons for low efficiency include, for example: hydrodynamic drag/turbulence losses around buoy, radiation of waves away from device, phase mismatch between buoy motion and excitation force, mechanical friction in PTO, generator copper/iron losses, power electronics conversion losses, cable I^2R losses, non-optimal operation in changing sea states.

One justified improvement: adaptive control (or adjustable damping/stiffness) to maintain favourable phase and amplitude response across varying periods, increasing time-averaged captured power.

(g) Suggested high-quality response points

$T_0 = 7.0\,\text{s}$ lies near the middle of 5.5 s to 9.0 s, so it is a compromise rather than an exact match to all conditions. Compare frequency mismatch at extremes:

$$f(5.5\,s) = 0.182\,\text{Hz}, \quad f(9.0\,s) = 0.111\,\text{Hz}, \quad f_0 = 0.143\,\text{Hz}.$$

The short-period case is farther from resonance than mid-range conditions; long-period swells are also offset. A fixed T_0 therefore gives acceptable but non-maximal capture over the full day. Candidates may argue for tunable systems (ballast/reactive control) to shift effective natural period, trading complexity and cost against higher annual energy yield.

Mark Scheme (CIE 9702 P4 style, Variant 2)

Q18(a) [4]

M1 for $f = 1/T$; A1 0.139 Hz. M1 for wave relation ($\lambda = v/f$ or $\lambda = vT$); A1 82.8 m.

Q18(b) [6]

M1 substitution into $P_{\text{wave}} = \frac{1}{8}\rho g A^2 v$; A1 $1.67 \times 10^5\,\text{Wm}^{-1}$. M1 use of interception width d; A1 $P_{\text{inc}} \approx 1.17 \times 10^6\,\text{W}$. M1 efficiency expression $\eta = P_{\text{elec}}/P_{\text{inc}}$; A1 $\eta \approx 5.3\%$.

Q18(c) [7]

M1 $\omega = 2\pi/T$; A1 $0.873\,\text{rads}^{-1}$. M1 $v_{\max} = \omega A$; A1 $2.97\,\text{ms}^{-1}$. M1 $a_{\max} = \omega^2 A$; A1 $2.59\,\text{ms}^{-2}$. M1 $KE_{\max} = \frac{1}{2}mv_{\max}^2$ with consistent substitution; A1 1.2×10^5 J (allow 1.20×10^5 J to 1.25×10^5 J).

Q18(d) [7]

M1 compute f_d; A1 0.1389 Hz. M1 compute f_0; A1 0.1429 Hz. A1 quantitative closeness statement (e.g. 0.004 Hz or about 3%). B2 explanation of risk near resonance (large amplitude, stress/fatigue/PTO overload/instability), max 2 for clear linked physics.

Q18(e) [7]

M1 $I = P/V$; A1 17.2 A. M1 $P_{\text{loss}} = I^2R$; A1 5.34×10^2 W. M1 percentage-loss method; A1 0.86% (allow 0.8–0.9%). B1 one valid engineering measure with correct reason (e.g. higher V, lower I, lower I^2R loss).

Q18(f) [9] Open-ended

Level-based marking with indicative content.

Level 3 (7–9): coherent, logically structured explanation of full energy pathway, at least three correct distinct loss mechanisms, and one justified improvement linked to physics

of capture/control.

Level 2 (4–6): partial pathway and at least two valid losses; improvement stated but weakly justified.

Level 1 (1–3): limited statements with minimal linkage; one valid idea only.

0: no creditworthy physics.

Q18(g) [10] Open-ended evaluation

Level 3 (8–10): balanced evaluation, quantitative comparison using given period range and T_0, correct resonance reasoning, clear judgment with trade-off (performance vs complexity/cost).

Level 2 (4–7): some quantitative content and relevant resonance discussion, but incomplete evaluation.

Level 1 (1–3): descriptive comments with little quantitative or evaluative support.

0: no creditworthy response.